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Safety data sheet

BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)
Date / Revised: 05.09.2024 Version: 3.0
Product: **Mixture of Catalysts (R210_R215)**
(ID no. 30720403/SDS_GEN_00/EN)
Date of print 01.07.2025

1. Identification

Product identifier

Mixture of Catalysts (R210_R215)

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Intermediate
Recommended use: Intermediate (isolated and transported)

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (Inhalation - dust)
Acute Tox. 5 (oral)
Eye Irrit. 2A
Resp. Sens. 1
Skin Sens. 1
Muta. 2
Carc. 1B
Repr. Additional category for effects on or via lactation.
Repr. 1B (fertility)
Repr. 2 (unborn child)
STOT SE 3 (irritating to respiratory system)
STOT RE (Respiratory system) 2 (by inhalation)
Aquatic Acute 1
Aquatic Chronic 2

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H303	May be harmful if swallowed.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317	May cause an allergic skin reaction.
H362	May cause harm to breast-fed children.
H335	May cause respiratory irritation.
H350	May cause cancer.
H341	Suspected of causing genetic defects.
H360	May damage fertility. Suspected of damaging the unborn child.
H373	May cause damage to organs (Respiratory system) through prolonged or repeated exposure (inhalation).
H411	Toxic to aquatic life with long lasting effects.
H400	Very toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe dust/gas/mist/vapours.
P201	Obtain special instructions before use.
P273	Avoid release to the environment.
P280	Wear eye protection.
P202	Do not handle until all safety precautions have been read and understood.
P284	In case of inadequate ventilation wear respiratory protection.
P272	Contaminated work clothing should not be allowed out of the workplace.
P263	Avoid contact during pregnancy and while nursing.
P264	Wash contaminated body parts thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P308 + P313	IF exposed or concerned: Get medical attention.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P391	Collect spillage.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 31 - 61 %, dermal
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 6 - 15 %, oral
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 46 - 91 %, Inhalation - dust

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

See section 12 - Results of PBT and vPvB assessment.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Preparation based on: metal oxides

Hazardous ingredients (GHS)
According to UN GHS criteria

Cobalt molybdate

Content (W/W): >= 16 % - <= 31,5 % CAS Number: 13762-14-6 EC-Number: 237-358-4 REACH registration number: 01-2119948547-24	Acute Tox. 5 (oral) Resp. Sens. 1 Skin Sens. 1 Muta. 2 Carc. 1B (by inhalation) Repr. 1B (fertility) Repr. 2 (unborn child) Aquatic Acute 2 Aquatic Chronic 2 H303, H334, H317, H341, H350, H360, H401, H411 Specific concentration limit: Carc. 1B: >= 0,01 %

Diiron trimolybdenum dodecaoxide

Content (W/W): >= 11,2 % - <= 23,8 % CAS Number: 13769-81-8 EC-Number: 237-389-3	Acute Tox. 5 (oral) H303

Molybdenum trioxide

Content (W/W): >= 8,3 % - <= 21,5 % CAS Number: 1313-27-5 EC-Number: 215-204-7 INDEX-Number: 042-001-00-9	Acute Tox. 5 (oral) Carc. 2 Eye Irrit. 2A STOT SE 3 (irr. to respiratory syst.) H319, H303, H335, H351

Dibismuth trioxide

Content (W/W): >= 2,3 % - <= 4,9 % CAS Number: 1304-76-3 EC-Number: 215-134-7	Acute Tox. 5 (oral) H303

Tricobalt tetraoxide

Content (W/W): >= 0,8 % - <= 1,7 % CAS Number: 1308-06-1 EC-Number: 215-157-2	Resp. Sens. 1B Aquatic Chronic 3 H334, H412

Divanadium pentaoxide

Content (W/W): >= 0,6 % - < 1,3 % CAS Number: 1314-62-1 EC-Number: 215-239-8 INDEX-Number: 023-001-00-8	Acute Tox. 2 (Inhalation - dust) Acute Tox. 3 (oral) Eye Dam./Irrit. 1 Muta. 2 Carc. 1B Repr. Add. cat. lact. Repr. 2 (fertility) Repr. 2 (unborn child) STOT SE 3 (irr. to respiratory syst.) STOT RE (Respiratory system) 1 (by inhalation) Aquatic Chronic 2 H318, H330, H301, H362, H335, H350, H341, H361, H372, H411

Copper oxide

Content (W/W): >= 0,2 % - < 0,9 % CAS Number: 1317-38-0 EC-Number: 215-269-1 INDEX-Number: 029-016-00-6	Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 100 M-factor chronic: 10 H400, H410

Tungsten trioxide

Content (W/W): >= 5,2 % - <= 10,9 %
CAS Number: 1314-35-8
EC-Number: 215-231-4

Graphite

Content (W/W): >= 1,6 % - <= 3,5 %
CAS Number: 7782-42-5
EC-Number: 231-955-3
REACH registration number: 01-2119486977-12

Silicon dioxide

Content (W/W): >= 1,2 % - <= 2,6 %
CAS Number: 7631-86-9
EC-Number: 231-545-4

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:

water jet, carbon dioxide

Additional information:
Use extinguishing measures to suit surroundings.

If a metal fire occurs, use dry sand, dry powder for Class D or cement.

Special hazards arising from the substance or mixture

Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings.

6. Accidental Release Measures

Shut off or stop released substance/product under safe conditions. Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air).

Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame. Use antistatic tools.

Environmental precautions

Discharge into the environment must be avoided.

Methods and material for containment and cleaning up

For small amounts: Dampen, pick up mechanically and dispose of.
For large amounts: Sweep/shovel up.

Collect waste in suitable containers, which can be labeled and sealed.

7. Handling and Storage

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice. Avoid all direct contact with the substance/product. Ensure thorough ventilation of stores and work areas. Change clothes immediately after contamination. Avoid the formation and deposition of dust.

Protection against fire and explosion:
The product is not an oxidizer, not self-combustible and not explosive. The substance/product is non-combustible.

Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container in a well-ventilated place. The authority permits and storage regulations must be observed.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

1308-06-1: Tricobalt tetraoxide
1313-27-5: Molybdenum trioxide
1314-35-8: Tungsten trioxide
1314-62-1: Divanadium pentaoxide
1317-38-0: Copper oxide
7631-86-9: Silicon dioxide
7782-42-5: Graphite
13769-81-8: Diiron trimolybdenum dodecaoxide
13762-14-6: Cobalt molybdate
14808-60-7: Quartz (SiO_2)
63210-56-0: Steatite ($Mg(SiO_3)$)

Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is required additionally to the stated personal protection equipment.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	solid
Form:	solid
Colour:	various colours grey to brown to black
Odour:	odourless
Melting temperature:	not applicable
boiling temperature:	not applicable
Lower explosion limit:	For solids not relevant for classification and labelling.
Upper explosion limit:	For solids not relevant for classification and labelling.
Flash point:	not applicable
Thermal decomposition:	Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. No decomposition if correctly stored and handled.
SADT:	Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.
pH value:	approx. 4,6 (20 °C)
Viscosity, kinematic:	not applicable, the product is a solid
Viscosity, dynamic:	not applicable, the product is a solid
Thixotropy:	not applicable
Solubility in water:	immiscible
Vapour pressure:	(20 °C) negligible
Relative vapour density (air):	The product is a non-volatile solid.

9.2. Other information

Information with regard to physical hazard classes

Explosives
Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Oxidizing properties
Fire promoting properties: not fire-propagating

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Bulk density: approx. 1.100 kg/m3
SAPT-Temperature:
Product does not fulfil criteria for polymerizing substances according to transport regulations.

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
None known during use and storage if used according to instructions.

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after short-term inhalation. Of low toxicity after single ingestion. The product has not been tested. The statement has been derived from the properties of the individual components.

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 31 - 61 %

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 6 - 15 %

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 46 - 91 %

Irritation

Assessment of irritating effects:
Eye contact causes irritation. The product has not been tested. The statement has been derived from the properties of the individual components.

Respiratory/Skin sensitization

Assessment of sensitization:
The substance may cause sensitization of the respiratory tract. Sensitization after skin contact possible. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment of sensitization:
Sensitization after skin contact possible. The substance may cause sensitization of the respiratory tract. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenic properties can not be excluded on the basis of experimental data. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was mutagenic in a mammalian cell culture test system. The substance was mutagenic in studies with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
The substance caused cancer in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Cobalt molybdate
Assessment of carcinogenicity:
The substance caused cancer in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
Causes impairment of fertility in laboratory animals. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment of reproduction toxicity:
Causes impairment of fertility in laboratory animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies. May cause harm to children via breast-feeding. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment of teratogenicity:
The results of animal studies gave indication of a developmental toxic effects. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Divanadium pentaoxide
Assessment of repeated dose toxicity:
The substance may cause damage to the lung even after repeated inhalation of low doses, as shown in animal studies. Repeated inhalation exposure to small quantities may affect certain organs.

Information on: Molybdenum trioxide
Assessment of repeated dose toxicity:
The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies.

Aspiration hazard

No aspiration hazard expected.

The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life. Toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not applicable for inorganic substances.

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Cobalt molybdate
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is possible.The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Dispose of in accordance with national, state and local regulations.
Used catalysts may have different hazardous properties than the original products.

Contaminated packaging:
Disposal must be made according to official regulations.

14. Transport Information

Land transport

ADR

UN number or ID number:	UN3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

RID

UN number or ID number:	UN3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport
ADN

UN number or ID number:	UN3077

UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-A; S-F

Air transport

IATA/ICAO

UN number or ID number:	UN 3077
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Eye Irrit.	Eye irritation
Resp. Sens.	Respiratory sensitization
Skin Sens.	Skin sensitization
Muta.	Germ cell mutagenicity
Carc.	Carcinogenicity
Repr.	Reproductive toxicity
STOT SE	Specific target organ toxicity — single exposure
STOT RE	Specific target organ toxicity — repeated exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Eye Dam./Irrit.	Serious eye damage/eye irritation
H303	May be harmful if swallowed.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317	May cause an allergic skin reaction.
H341	Suspected of causing genetic defects.
H350	May cause cancer by inhalation.
H360	May damage fertility. Suspected of damaging the unborn child.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H319	Causes serious eye irritation.
H335	May cause respiratory irritation.
H351	Suspected of causing cancer.
H412	Harmful to aquatic life with long lasting effects.
H318	Causes serious eye damage.
H330	Fatal if inhaled.
H301	Toxic if swallowed.
H362	May cause harm to breast-fed children.
H350	May cause cancer.
H361	Suspected of damaging fertility. Suspected of damaging the unborn child.
H372	Causes damage to organs (Respiratory system) through prolonged or repeated exposure (inhalation).
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.